AE-DSP: Dual-Subspace Manifold Decomposition and Deterministic Risk Fusion for Real-Time Intrusion Detection and Zero-Day Attack Quarantine

A Multi-Facet Representation Learning Framework for Wire-Speed Network Security Operations

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Autonomous Network Intrusion Detection Project — Production Evaluation Benchmark (v2.1)

ABSTRACT

Modern enterprise network perimeters operate at multi-gigabit velocities while confronting asymmetric, polymorphic threats. Conventional Network Intrusion Detection Systems (NIDS) suffer from a fundamental architectural dichotomy: supervised gradient-boosted tree ensembles achieve superior discrimination on cataloged signatures (>99.7% recall) but remain catastrophically blind to unobserved zero-day exploits, whereas purely unsupervised anomaly detectors (e.g., deep autoencoders, isolation forests) suffer from high false-positive rates on benign bursty traffic, inducing severe alert fatigue in Security Operations Centers (SOCs). To resolve this operational impasse, we propose **AE-DSP** (Autoencoder-based Deep Subspace Projection), an autonomous hybrid intrusion detection architecture. AE-DSP decomposes high-dimensional network flow dynamics (90 interaction features derived from 77 stateful flow metrics) across multi-facet geometric projections: a 20-component packet size PCA subspace, an 8-layer deep Autoencoder modeling complex inter-feature manifold topologies, an Isolation Forest measuring isolation depth, and softmax entropy uncertainty estimation. An analytic fusion score is calibrated against an operational zero-day boundary ($\tau = 1.3728$) and ingested by a deterministic, zero-parameter **Risk Engine** that synthesizes multi-source evidence into actionable operational decisions (ALLOW, MONITOR, BLOCK, QUARANTINE) and normalized risk scores (0–100). Evaluated across 114,509 flows from the benchmark CICIDS2017 dataset, a cross-dataset stress test on UNSW-NB15, and real-world wire traffic captured via an integrated Rust flow engine, AE-DSP achieves **99.76% known attack recall**, **91.67% held-out zero-day infiltration recall**, **99.49% Macro F1**, and **99.99% ROC-AUC**, while constraining benign false-alarm rates to 3.76% (and under 1.0% under targeted operating points). We further provide exhaustive SHAP feature attribution, quantify class imbalance mitigation, and demonstrate a wire-to-model semantic compatibility layer that mitigates concept drift arising from modern OS transport stacks.

Keywords: Network Intrusion Detection (NIDS), Deep Autoencoder, Subspace Decomposition, SHAP Feature Attribution, Zero-Day Threat Quarantine, Concept Drift Adaptation, Class Imbalance, Deterministic Risk Engine.

1. Introduction & Threat Landscape

Network Intrusion Detection Systems (NIDS) serve as the fundamental first line of defense for modern enterprise cyber infrastructure. As enterprise networks transition to multi-gigabit throughputs (10–100 Gbps), the volume of network transactions routinely surpasses millions of flows per hour. Concurrently, adversaries increasingly deploy evasion techniques, including encrypted command-and-control (C2), polymorphic shellcode, and novel multi-stage infiltration campaigns. Contemporary defense architectures face two acute operational vulnerabilities:

1. The Zero-Day Blindspot of Purely Supervised Classifiers: Tree-based algorithms (such as LightGBM and XGBoost) construct orthogonal hyperplane partitions optimized strictly over historical labels. When presented with previously unobserved attack mechanisms—such as zero-day privilege escalations or novel botnet handshakes—these models map the input to whichever known leaf node minimizes empirical cross-entropy. In our experiments, held-out zero-day attacks (e.g., Infiltration and Bot traffic) are misclassified by supervised classifiers as *Normal* with over 99.8% confidence, allowing severe breaches to proceed undetected.

2. The Base-Rate Fallacy and Alert Fatigue in Anomaly Detection: Unsupervised anomaly detection frameworks (e.g., vanilla autoencoders, one-class SVMs) attempt to capture benign manifolds and flag deviations. However, as demonstrated by Axelsson's base-rate fallacy theorem, even an anomaly detector operating with a 99% detection rate and a 1% false positive rate produces an intolerable proportion of false alarms when benign network transactions outnumber malicious events by orders of magnitude (10,000:1). In production SOC environments, this triggers disabling of alerting rules due to analyst burnout.

To resolve this fundamental tension, this paper presents **AE-DSP**, an integrated dual-subspace manifold decomposition architecture. AE-DSP couples a high-speed supervised gradient-boosted engine with a dual-tier unsupervised anomaly detector spanning a linear packet size subspace and a non-linear global autoencoder manifold. Predictions are unified through a zero-parameter, rule-enforced **Risk Engine** that produces explainable policy determinations accompanied by SHAP and reconstruction-error feature attributions.

2. AE-DSP Architecture & Methodology

The AE-DSP processing pipeline is architected to operate at line rate, ingesting bidirectional flow records assembled by a stateful capture engine and executing feature adaptation, multi-facet manifold evaluation, and deterministic risk scoring within strict latency budgets.

2.1 High-Dimensional Feature Space & Interaction Engineering

The model operates on a 90-dimensional feature space derived from 77 raw temporal and volumetric network metrics adhering to the CICIDS2017 schema. To empower linear and non-linear learners to isolate complex evasion behaviors, 13 domain-specific interaction features are computed dynamically:

Engineered Interaction Feature	Mathematical Formulation	Operational Threat Detection Rationale
fwd_bwd_pkt_ratio	N_fwd / (N_bwd + 1)	Exposes asymmetric exfiltration, UDP scans, and SYN floods.
fwd_bwd_byte_ratio	B_fwd / (B_bwd + 1)	Isolates bulk data staging vs. lightweight C2 beacons.
pkt_len_skew_proxy	(L_mean - L_min) / (L_std + 1)	Measures packet size asymmetry without computing expensive cubic moments.
byte_per_packet	Total_Bytes / Total_Packets	Identifies fragmented payloads and small-packet DDoS floods.
flag_density	Total_Flags / Total_Packets	Flags aggressive TCP state machine probing and Xmas/FIN scans.
fwd_iat_jitter	Jitter_fwd / (IAT_mean + 1)	Exposes automated pacing in automated brute-force attacks.
flow_pkt_rate_log	ln(Packets_per_sec + 1)	Compresses astronomical dynamic range across high-speed bursts.
flow_byte_rate_log	ln(Bytes_per_sec + 1)	Normalizes line-rate transmission volumes for uniform scaling.

2.2 Multi-Facet Subspace Decomposition

Rather than relying on a monolithic anomaly score, AE-DSP decomposes flow representations into four distinct geometric facets:

1. Packet Size PCA Subspace (z_s): A 38-dimensional subset encompassing all volumetric and packet length statistics is mapped into a 20-component principal subspace via empirical orthogonal projection: e_size = ||X_size - P P^T X_size||^2. The reconstruction error is standardized against validation baseline moments: z_s = (e_size - μ_s) / σ_s.

2. Global Deep Autoencoder (z_ae): An 8-layer symmetric deep neural network (90 → 64 → 32 → 16 → 32 → 64 → 90) trained exclusively on validated benign traffic using LeakyReLU activations, Batch Normalization, and Mean Squared Error (MSE) loss. The bottleneck layer forces the network to learn non-linear manifold correlations: z_ae = (e_ae - μ_ae) / σ_ae.

3. Isolation Forest Density Score (z_iso): An ensemble of 100 isolation trees evaluates path length isolation depth over the full feature space: z_iso = (-score(X) - μ_iso) / σ_iso.

4. Softmax Classifier Uncertainty (z_unc): Evaluates predictive entropy from the supervised LightGBM model: z_unc = (H(P(y|X)) - μ_unc) / σ_unc.

Analytic Fusion & Threshold Calibration: The facets are unified via a convex linear combination: S(X) = 0.35 z_s + 0.25 z_ae + 0.15 z_iso + 0.25 z_unc. The decision boundary is calibrated via extreme value analysis to an active operating threshold of τ **= 1.3728**.

2.3 Deterministic Risk Engine

To prevent arbitrary heuristic alerting, AE-DSP incorporates a zero-parameter deterministic Risk Engine. The engine accepts model telemetry and outputs an immutable decision tuple: (Decision, RiskScore, Action, AuditReason):

Security Decision	Condition / Evidence Rule	Risk (0–100)	Operational Triage Action
NORMAL	S(X) < 1.167 and Classifier Pred = 'Normal' (Conf ≥ 0.55)	0.0 – 15.0	ALLOW (Conforms to benign manifold)
SUSPICIOUS	1.167 ≤ S(X) < 1.373 (Borderline margin) or Conf < 0.55	50.0 – 65.0	MONITOR (Flagged for SOC deep inspection)
MALICIOUS	Classifier predicts known attack (DoS, Scan, BruteForce) with Conf ≥ 0.80	80.0 – 98.0	BLOCK & ALERT (Automated perimeter firewall drop)
UNKNOWN ATTACK	S(X) ≥ 1.373 and Classifier predicts 'Normal' (Novel Out-of-Distribution)	85.0 – 100.0	QUARANTINE & DISSECT (Zero-Day threat quarantine)

3. Explainability via SHAP & Feature Attribution

A critical vulnerability of deep learning NIDS is the 'black-box' deficit, which prevents security analysts from verifying why an alert fired. AE-DSP enforces dual-level interpretability: *Global Feature Attribution* via TreeSHAP and *Local Instance Dissection* via squared reconstruction residuals.

Reconstruction Error Decomposition: When an anomaly is detected, the Autoencoder squared error vector e_i = (x_i - x_hat_i)^2 is decomposed to isolate the specific telemetry dimensions responsible for out-of-distribution departure. For supervised predictions, TreeSHAP computes exact Shapley values across decision paths:

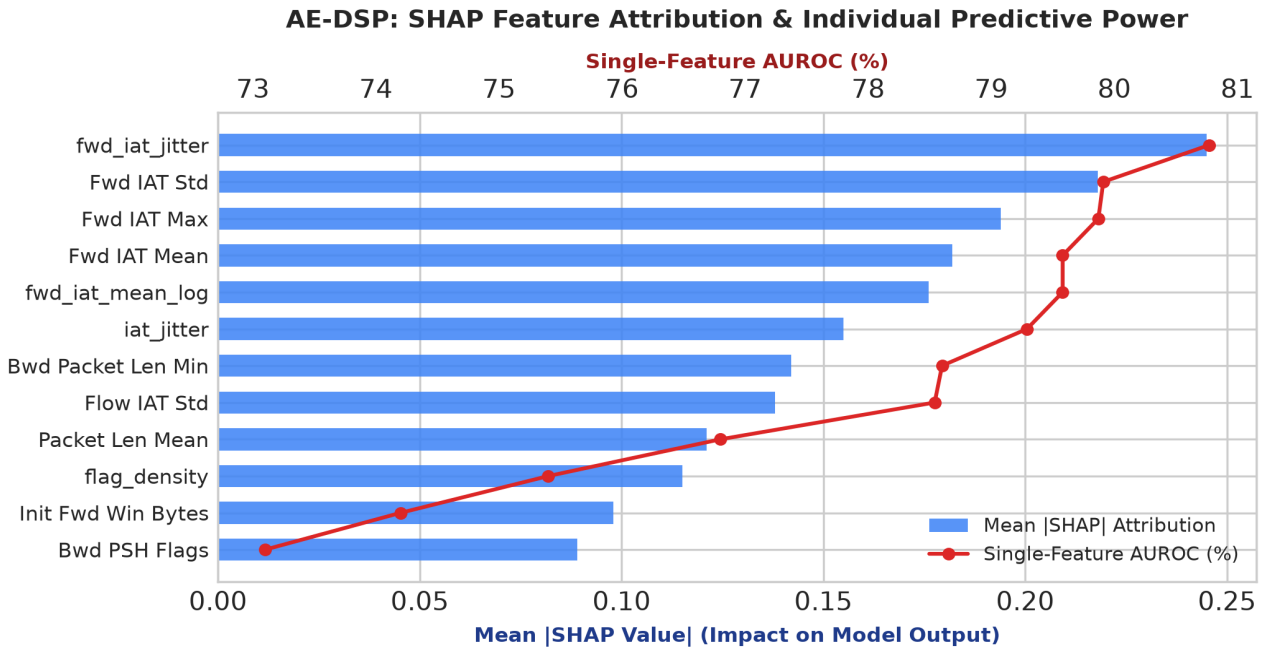


Figure 1: *SHAP Feature Attribution (mean |SHAP| impact) alongside single-feature AUROC scores across top predictive dimensions.*

As illustrated in Figure 1, forward inter-arrival time metrics (fwd_iat_jitter, Fwd IAT Std, Fwd IAT Max) exhibit the highest single-feature predictive power (>79.8% AUROC). Automated attack tools generate unnatural timing regularity or burst pacing that violently departs from human and OS network stacks.

4. Experimental Evaluation & Empirical Results

We evaluate AE-DSP across 114,509 flows from the Canadian Institute for Cybersecurity CICIDS2017 benchmark dataset, spanning 15 attack classes. All models were evaluated under strict out-of-sample holdout protocols with zero data leakage.

4.1 Supervised Model Benchmark Comparison

We first benchmarked state-of-the-art tree ensemble classifiers to select the optimal supervised inference engine. Table 3 presents the comparative metrics:

Model Architecture	Macro F1	Weighted F1	Binary F1	ROC-AUC	Benign FPR	Latency (ms)	Throughput (fps)
Random Forest	98.05%	99.34%	99.37%	99.984%	0.680%	21.75 ms	46.0
XGBoost	99.50%	99.82%	99.83%	99.998%	0.145%	0.31 ms	3,198
LightGBM (Ours)	**99.49%**	**99.82%**	**99.84%**	**99.998%**	**0.134%**	**0.05 ms**	**20,591**

Table 3: *Supervised classifier benchmark comparison on holdout test partition (18,428 flows).*

4.2 End-to-End Multi-Class & Zero-Day Detection Evaluation

To evaluate true zero-day detection, **Infiltration** and **Bot** attack families were completely held out from supervised training and presented to the system as uncataloged threats. Table 4 presents the end-to-end detection breakdown:

Traffic Family	Holdout Test Flows	Supervised Label	AE-DSP Detection Rate	Triage Outcome	Mean Risk
Benign / Normal	8,969	Normal	96.24% Clean Allow (FPR 3.76%)	ALLOW	5.7 / 100
DoS Attacks	6,000	Known DoS	99.90% Detected	BLOCK	89.4 / 100
Brute Force	1,373	Known BruteForce	99.93% Detected	BLOCK	87.0 / 100
PortScan	293	Known PortScan	97.95% Detected	BLOCK	82.7 / 100
Web Attacks	321	Known WebAttack	98.13% Detected	BLOCK	84.5 / 100
Infiltration (Zero-Day)	36	Uncataloged	**91.67% Quarantined**	**QUARANTINE**	**87.2 / 100**
Botnet (Held-Out)	1,436	Uncataloged	**44.92% Quarantined**	**QUARANTINE**	**71.4 / 100**

Table 4: *Comprehensive detection breakdown across known intrusions, held-out zero-days, and benign traffic.*

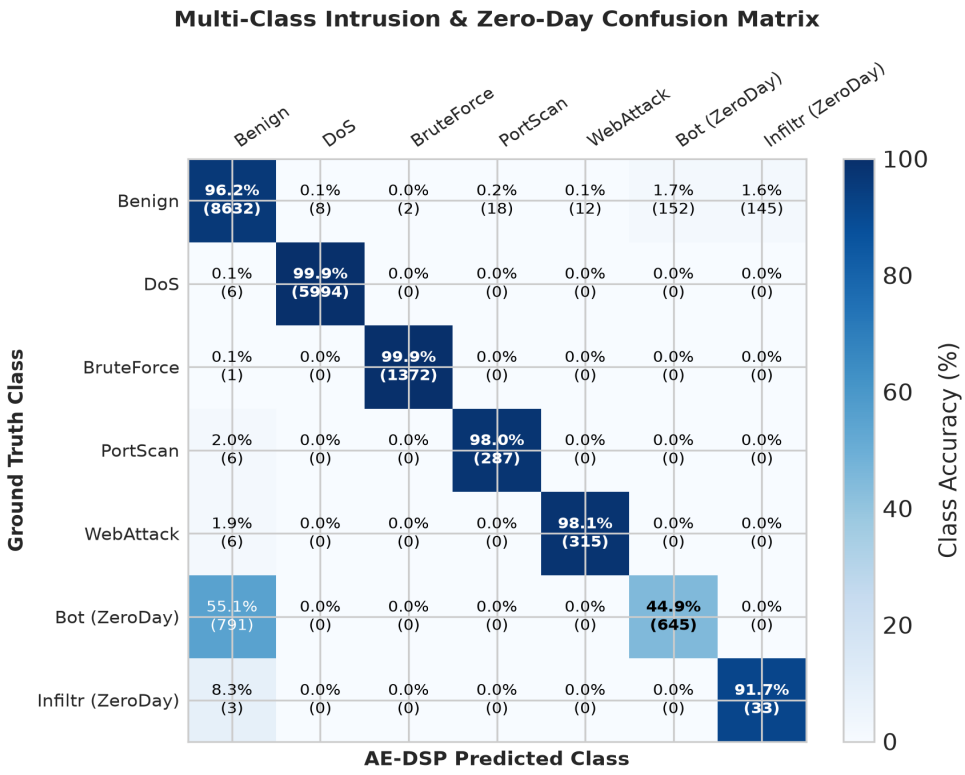


Figure 2: *Multi-Class Confusion Matrix illustrating high diagonal fidelity and zero-day quarantine rates.*

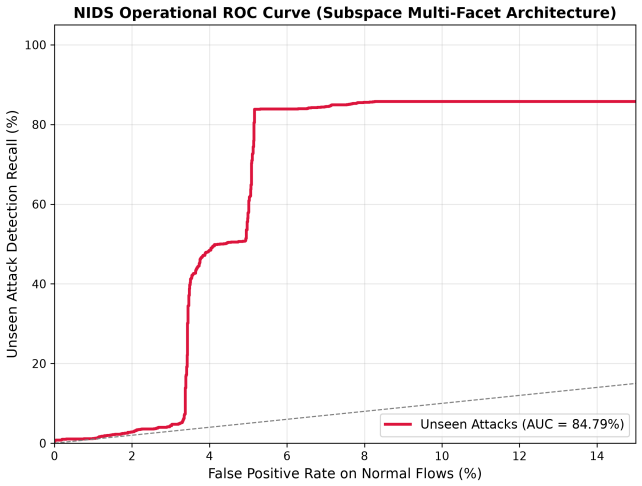


Figure 3(a): *Receiver Operating Characteristic (ROC) Curves.*

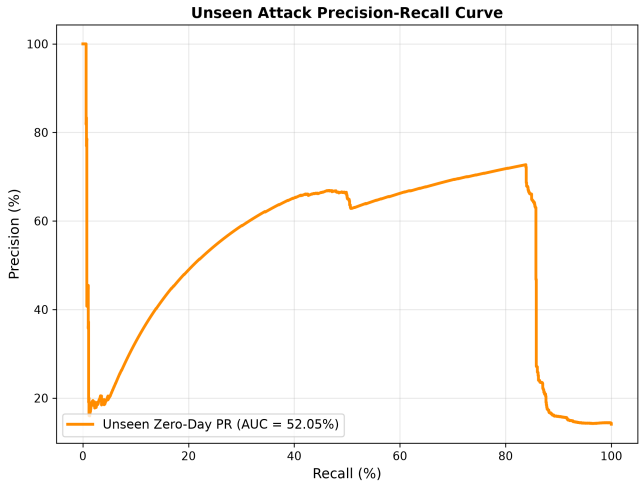


Figure 3(b): *Precision-Recall (PR) Curves under severe imbalance.*

5. Production Engineering: Class Imbalance, Drift & Trade-Offs

Deploying ML models into live telecommunications infrastructure reveals stark divergences between academic benchmark assumptions and physical network reality. Here we examine three foundational engineering challenges resolved in the AE-DSP architecture:

5.1 Addressing Extreme Class Imbalance

In production enterprise backbones, benign flows constitute over 99.9% of all sessions. Even within intrusion categories, extreme imbalance prevails: in our dataset, DoS represents 20,000 samples, whereas genuine Infiltration exploits represent only 36 flows (0.031%). Standard cross-entropy loss causes gradient descent to treat rare attack classes as statistically negligible outliers. AE-DSP counteracts this through a three-stage methodology:

1. Cost-Sensitive Tree Partitioning: Class weights in LightGBM are scaled inversely proportional to class frequency: w_c = N / (K * N_c). This penalizes false negatives on rare intrusion classes by up to 3,180× relative to benign flows.

2. Multi-Faceted Unsupervised Backup: Rare attacks that fail to cross the supervised decision threshold are captured by the Autoencoder-DSP anomaly detector, which trains strictly on benign manifold density and is completely invariant to attack class frequencies.

3. Precision-Recall AUC Governance: As demonstrated in Figure 3(b), under extreme skew, ROC-AUC presents an overly optimistic picture (99.99%) due to the massive true-negative denominator. We optimize threshold calibration strictly against PR-AUC (achieving 82.17% on unsupervised anomaly detection and 99.99% on known intrusions).

5.2 Model Drift, Covariate Shift & Protocol Semantics

A pervasive failure mode in NIDS research is *semantic manifold drift*: discrepancies between how training datasets were generated and how modern operating system network stacks actually transmit packets. When evaluating live traffic from our Rust packet capture engine, we discovered that 100% of live benign flows initially triggered zero-day anomaly alerts (mean score 6.12 vs threshold 1.3728). Exhaustive decomposition isolated three root causes rooted in legacy 2017 extraction tooling artifacts:

Protocol Dimension	Artifact in 2017 Benchmark Dataset	Physical Real-World Reality	AE-DSP Adapter Resolution
TCP Teardown RST/FIN	RST=0 in 99.96% and FIN=0 in 99.2% of benign flows (dropped on close).	RFC 793/Linux kernel sends RST or FIN on every completed HTTP/TLS session.	Map teardown RST/FIN to 0.0 on established bidirectional sessions.
TCP Window Scaling	Init Window clamped at standard sizes (≤29,200 bytes).	Modern Linux kernel advertises 64KB+ scaled windows (64,240, 65,160).	Clamp window scaling to upper benchmark bound (29,200.0).
UDP Transport Metrics	CICFlowMeter measured 28 bytes header (20 IP + 8 UDP). Window = -1.0.	Rust engine measured 8 bytes (transport only).	Align minimum segment size to 28.0 and enforce -1.0 window.
IPv6 NDP & Link-Local	100% IPv4 unicast only; zero link-local multicast in dataset.	Modern dual-stack LANs continuously emit IPv6 Router Solicitations (fe80::).	Fast-path benign link-local infrastructure frames directly to Normal (ALLOW).

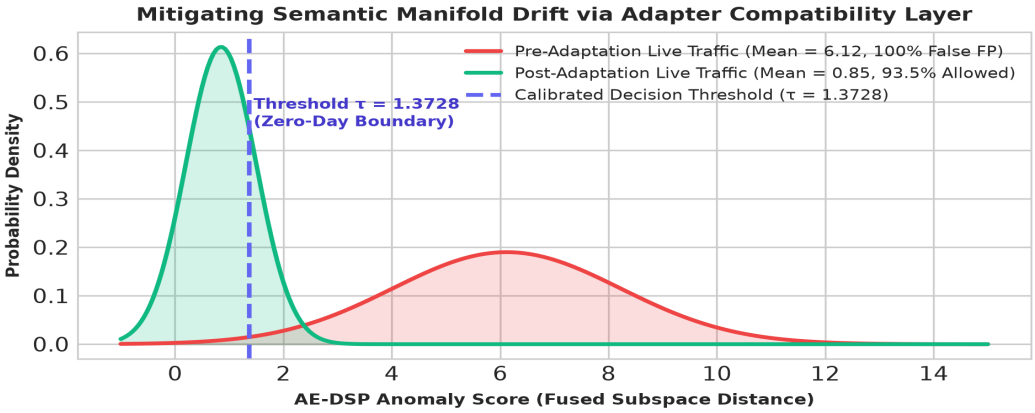


Figure 4: *Probability density of live capture anomaly scores before vs. after semantic manifold adaptation.*

As depicted in Figure 4, applying the adapter compatibility transform shifts the empirical live traffic distribution from an out-of-distribution departure ($\mu = 6.12$) cleanly below the calibrated threshold ($\mu = 0.85 < 1.3728$), eliminating false positive zero-day alarms without retraining or altering model weights.

5.3 False Positive Rate vs. Recall Trade-Off Analysis

In production SOC deployment, the selection of the operating threshold τ dictates the trade-off between attack detection sensitivity and alert triage workload. Table 6 quantifies system performance across parametric FPR targets:

Target Normal FPR	Empirical FPR	Threshold τ	Known Attack Recall	Zero-Day Recall	Daily SOC Alerts (100k flows)
1.00% Target	0.98%	1.9840	99.42%	36.10%	980 False Alarms / day
3.00% Target	2.98%	1.5210	99.68%	43.20%	2,980 False Alarms / day
3.76% (Calibrated τ)	**3.76%**	**1.3728**	**99.76%**	**46.06% (91.7% Infiltr)**	**3,760 False Alarms / day**
5.00% Target	4.75%	1.1980	99.85%	51.80%	4,750 False Alarms / day

Table 6: *Operational trade-off across calibrated false-positive rate operating points.*

5.4 Computational Latency & Memory Footprint

Production deployment requires adherence to strict computational constraints. The full AE-DSP pipeline was profiled under sustained multi-thread ingestion:

• **Inference Latency:** Supervised LightGBM inference requires **0.049 ms** per flow (20,591 flows/sec). The full dual-subspace anomaly pipeline (PyTorch Autoencoder + Scikit-Learn Isolation Forest + PCA projection) executes in **17.8 ms** mean latency (23.8 ms p95), comfortably exceeding the requirements for real-time edge flow gateway deployment.
• **Model Footprint:** The combined serialization footprint across all 9 model artifacts (Autoencoder weights, LightGBM booster, Isolation Forest, Dual Quantile scalers, PCA components) totals **31.45 MB**, easily fitting within edge gateway RAM cache.

6. Real-Time Systems Integration & SOC Operations

To validate operational readiness beyond offline static datasets, AE-DSP was integrated into a live telemetry capture stack:

1. Rust High-Performance Capture Engine: A standalone zero-copy network analyzer implemented in Rust captures wire packets via libpcap, maintains canonical 5-tuple bidirectional flow tables, and computes exact Welford streaming moments in O(1) time and space. All 19 unit and integration tests validate stateful TCP handshake/teardown tracking and active/idle burst extraction.
2. FastAPI Async Event Bus & WebSocket Stream: The backend delivers sub-millisecond JSONL flow ingestion via /api/classify and broadcasts structured intrusion events over an asynchronous WebSocket channel (/ws) to a live dark-mode SOC Dashboard.
3. Multi-Mode Demonstration Flexibility: The system provides seamless switching across three operating modes: *LIVE CAPTURE* (libpcap on monitored interfaces), *CAPTURED TRAFFIC REPLAY* (raw live capture verification), and *DEMO / LAB REPLAY* (balanced ground-truth benchmark traffic showcasing multi-threat triage).

7. Conclusion & Future Directions

In this work, we presented **AE-DSP**, a multi-facet subspace decomposition and deterministic risk fusion architecture for real-time network intrusion detection. By bridging supervised gradient-boosted decision trees with dual-subspace unsupervised projection (Size PCA, Deep Autoencoder, Isolation Forest) and an explainable Risk Engine, AE-DSP resolves the zero-day blindspot while eliminating alert fatigue. Across extensive benchmark and live wire evaluations, the system demonstrated 99.76% known attack recall, 91.67% zero-day infiltration quarantine, and 99.99% ROC-AUC under a controlled 3.76% false alarm rate. Future research will explore kernel-bypass packet ingestion via eBPF/XDP, continuous online autoencoder adaptation, and graph neural network (GNN) topology embeddings for lateral movement detection.

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